

LESSON GUIDE

Technology Plus

Unit 4: Networking

2.8.1 – Basic Network Concepts

Lesson Overview:

In this lesson, students will identify key components of a computer network and explain how devices communicate using protocols, addresses, and ports.

Students will:

- Recognize how protocols, IP addresses, and ports are used to send and receive data on a network.
- Identify the function of common networking devices and services.
- Use built-in network tools to examine a device's network configuration and connectivity.

Guiding Question: How do devices know where to send data on a network, and what makes that communication work?

Suggested Grade Levels: 8-12

Technology Needed: Yes

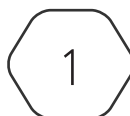
Approximate Time Required: 75-90 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.8 – Identify basic networking concepts.
 - Basics of Network Communication
 - Network identifiers
 - IP address
 - Media access control (MAC) address
 - Ports
 - Basic network services
 - Secure web browsing
 - File transfer
 - Email
 - Network devices
 - Modem
 - Router
 - Switch
 - Access point
 - Firewall
 - Networking models
 - Client/server
 - Peer-to-peer
 - Local area network (LAN)
 - Wide area network (WAN)

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Lesson Background

Background Information:

This lesson lays the foundation for nearly every other concept in networking, cybersecurity, and systems troubleshooting. Students won't walk away as network engineers, but they will start recognizing the invisible rules that govern how their devices connect, communicate, and fail. By introducing protocols, addresses, ports, and devices in a clear, structured way, this lesson gives students the language and mental model they need to make sense of what's happening when "the Wi-Fi's down" or when a tool like ping tells them more than just "it's broken."

It's also a soft launch into understanding how systems are secured – since everything from firewalls to DNS filtering builds on the assumption that data flows along predictable, inspectable paths. This is the moment when students stop seeing networks as magic and start seeing the map.

Materials Needed:

Materials per Class	- Lesson Guide 2.8.1 – Basic Network Concepts - Slides 2.8.1 – Basic Network Concepts
Materials per Student	- Student Notes 2.8.1 – Basic Network Concepts - Student Handout 2.8.1 – Basic Network Concepts - Lab Slides 2.8.1 – Know Your Network: IPs, Ports & Connectivity - Assessment 2.8.1 – Basic Network Concepts (optional)

Lesson Background

Cyber Terms:

Term	Definition
network	a group of connected devices – like computers, phones, or printers – that use cables or Wi-Fi to share information and resources
protocol	a set of rules that devices follow to send, receive, and understand data – like a shared language for computers
Internet Protocol (IP) address	a unique number that identifies a device on a network and helps data find the right destination, like a mailing address
media access control (MAC) address	a permanent, built-in identifier for a device's network connection – kind of like its fingerprint on a local network
port	a virtual doorway that helps data reach the right app or service on a device (like port 80 for websites or 587 for email)
modulator-demodulator (MODEM)	a device that connects a local network to the internet by converting digital signals into ones your ISP can use
router	a device that sends data between your local network and the internet; it also gives devices IP addresses and uses NAT to manage traffic
switch	a device that connects multiple devices in a local network and directs data to the right one using MAC addresses so they can communicate efficiently
access point	a device that lets wireless devices connect to a network, usually by adding Wi-Fi to a wired system
firewall	a security tool (hardware or software) that monitors and filters network traffic to decide what data gets in or out – like a digital bouncer protecting the network
Dynamic Host Configuration Protocol (DHCP)	a service that automatically gives IP addresses to devices on a network so they can connect without manual setup
Domain Name System (DNS)	the system that translates website names (like cyber.org) into IP addresses so your device knows where to go online
local area network (LAN)	a small network in one location, like a home, classroom, or office
wide area network (WAN)	a large network that connects devices over long distances – often made up of smaller LANs connected together
client/server	a network setup where one device (the server) provides data or services, and other devices (clients) request them
peer-to-peer	a network setup where all devices can share data directly with each other, without a central server

2.8.1 – Basic Networking Concepts

Guiding Question: How do devices know where to send data on a network, and what makes that communication work?

Lesson Launch (5 minutes)

- Read through the guiding question and the objectives.
- Then ask your students if they know how their phone knows how to get to YouTube. Ask them how their computer knows whether to load Google or send an email.
- Do a brief think-pair-share to have them discuss these questions or have a class-wide discussion with an open callout.
- Some students may initially respond that the phone or computer ‘just knows.’ Others with prior knowledge of networking and protocols may be able to identify key concepts and provide more advanced insights.
- Click to advance the animation and tell the students that today we are pulling back the curtain on how devices talk to each other - and how they know where to send data.

Networking: The Basics (30 minutes)

- Distribute the Student Handout 2.8.1 – Basic Networking Concepts and have students fill in the answers during the lesson and lab.

What Is a Network?

- Group of connected devices that share data and resources
- Devices = nodes
- Use protocols to communicate
- Example: The Internet
- This opening slide introduces the idea of a network using familiar devices. Emphasize that protocols are the “language” of networked devices, and that students interact with networks every day—even without realizing it.

Components of a Network

- **Network Media:** Ethernet, fiber optics, Wi-Fi
- **Network Interface:** NICs, wired/wireless
- **Network Protocols:** TCP/IP, HTTP, HTTPS, DNS
- Clarify the difference between physical and virtual components. NICs serve as the handshake between a device and the network. Highlight common student devices—phones, laptops, gaming consoles—as examples.

How Data Moves

Broken into **packets**

Follows **protocol rules**

- Devices send, receive, and route packets
- **Key Concepts:**
- **Bandwidth** = max speed
- **Latency** = delay in transmission
- This is a great moment to compare bandwidth to highway lanes and latency to traffic delays. Reinforce that more bandwidth $\neq$ less latency. Run a speed test demo if time allows.

IP Addressing (Logical Address)

- Internet Protocol = temporary, software-assigned
- **IPv4**: 32-bit, e.g. 192.168.1.1
- **IPv6**: 128-bit, hex groups
- Assigned via **DHCP** or manually
- **Subnetting**: Manageable chunks
- **Subnet Mask**: Separates network/device parts
- **Public vs. Private** IPs
- Use student devices in the lab to identify local vs. public IPs. Emphasize that IPs are like mailing addresses: they tell traffic where to go.

NAT (Network Address Translation)

- Lets many devices share one public IP
- Router does this silently
- Common in home & small office networks
- Use the hotel analogy—everyone has a room number (private IP), but the hotel shares one street address (public IP). The router is the front desk handling message delivery.

MAC Address (Physical Address)

- Permanent hardware ID
- Assigned at manufacturer
- Hexadecimal format
- Used by switches within LAN
- Contrast MAC vs IP: MAC = name tag burned into device, IP = mailing label that can change. MACs never leave the LAN.

What Are Ports?

- Decide which service handles data
- Example: Port 80 = web, 587 = email
- **Common Ports:**
- **21** FTP
- **22** SSH
- **53** DNS
- **80** HTTP
- **443** HTTPS
- **587** SMTP (secure)
- A port is like an apartment unit number—it tells the data where exactly to go.

Security Tip + Port Forwarding:

- Prefer secure ports: 443 > 80, 587 > 25
- Block unused ports to reduce attack risk
- **Port Forwarding:**
- Router redirects data to correct internal device
- Example: Port 3389 for remote desktop
- A real-world example is gaming consoles needing port forwarding. Students may recognize this from Xbox/PlayStation settings.

Basic Network Services

- **Secure Web Browsing:** HTTPS, SSL/TLS
- **File Transfer:** FTP (21), SFTP (22), FTPS
- **Email:** SMTP (587), IMAP (143/993), POP3 (110/995)
- **DNS:** Domain name to IP, port 53
- **DHCP:** Assigns IPs, ports 67 (server)/68 (client)
- Reinforce that these services run in the background constantly. Walk through sending an email or loading a website to show what's happening behind the scenes.

Networking Devices?

- **Modem:** Connects to ISP
- **Router:** Directs traffic, assigns IPs, uses NAT
- **Switch:** Sends data using MAC addresses
- **Access Point:** Wireless entry to LAN
- **Firewall:** Filters traffic
- **Bonus:** PoE Switches
- Power & data via one cable
- Used for: access points, IP cams, VoIP, smart devices
- If possible, show real or labeled images of each device. PoE is a nice advanced topic—students may have seen ceiling-mounted devices at school.

Network Models

- **Client/Server:**
 - Centralized service provider
 - Good control & security
- **Peer-to-Peer:**
 - Equal devices share data directly
 - Cheap but less secure
 - Compare to school cafeteria (client/server) vs. a potluck (P2P). Ask which model they think texting or multiplayer games use.

Network Types

- **LAN:** Small area, e.g., home
- **WAN:** Large area, city/country
- **MAN:** Metropolitan (city/campus)
- **WLAN:** Wireless LAN
- **PAN:** Personal (Bluetooth)
- Ask students to give examples from their lives - Bluetooth earbuds = PAN, school network = LAN.

LAN vs WAN

Feature	LAN	WAN
Area	Small	Large
Speed	High	Varies
Cost	Low	High
Ownership	Private	Often leased/shared
Admin	Local	Centralized/remote
Devices	Switch, Router, AP	Router, Modem, Firewall

- Emphasize that WANs are like the freeway system, while LANs are neighborhood streets. Ownership and cost are major distinctions.

LAN & VLAN

- **LAN:**
 - Small area
 - Uses switches, routers, APs
 - Admin controls access/security
- **VLAN:**
 - Logically splits physical LAN
 - Devices in different VLANs can be isolated
 - Great for security/performance
 - Use the “classroom seating chart” analogy—everyone’s in the same room but in different groups with different permissions.

WAN & Examples

- **WAN:**
 - Connects LANs across large regions
 - Managed by ISPs, orgs, or governments
- **Examples:**
 - Corporate WAN linking global offices
 - Cloud services (AWS, Azure, Google Cloud)
 - Clarify that WANs often use VPNs or leased lines. The internet is technically a giant public WAN.

Lab: 2.8.1 Know Your Network: IPs, Ports, & Connectivity (30-45 minutes)

Part 1: Get Your Network Info

- In this lab, students use built-in network tools to explore device configurations, open ports, and communication between systems - all within a secure cyber range environment.
- In real networks, MAC addresses can sometimes tell you what manufacturer made a device - especially on physical hardware. But in virtual labs like this one, MACs are often fake, random, or reused by the virtualization software, so they don't point to a real company.

Part 2: DNS Lookup Practice

- When you run nslookup on sites like facebook.com, you'll get a redirect to a placeholder address (like blocked.orgcyberrange.com) instead of the real IP.
- This range uses DNS filtering to block certain sites, just like schools and companies do to control access. It's a real-world example of how firewalls and content filters manage what traffic is allowed in and out of a network.

Part 3: What's Listening?

- The netstat (Windows) and ss (Ubuntu) commands show which services are "listening" on specific ports - this means the system is ready to receive connections on those ports.
- Every open port is tied to a specific service or protocol. For example, if port 80 is open, it usually means a web server is running. Port 22? That's SSH.
- On classroom devices or personal computers, you might not see many open ports. But on servers, you'd see ports for websites, email, remote access, etc.
- This part of the lab helps students connect port numbers (from their notes) to real services. It's also a great time to remind them **that open ports can be a security risk** – that's why administrators close unused ones.
- Bonus: In the Ubuntu section, students stop and restart the CUPS service to see port 631 disappear and return. That's a real-time example of how a service controls whether a port is open.

- When students restart the CUPS service, port 631 shows up again – but only as a TCP connection. That’s because:
 - CUPS primarily uses TCP, not UDP, to handle print jobs. TCP is better for tasks that need reliability and order (like sending documents to a printer).
 - The UDP connection they saw earlier may have been a temporary multicast or broadcast used for printer discovery, which doesn’t always restart with the main CUPS service.

Part 4: Ping Around the Range

- The ping command tests network connectivity by sending a small packet (called an ICMP Echo Request) to another device and waiting for a reply. If the device responds with an Echo Reply, it means the connection is working.
- Having a machine ping itself is the quickest way to confirm that its network software is working – even without a network connection. As a troubleshooting step, it helps determine whether the issue lies with the computer itself or with the network it’s trying to reach.
- Students will need to go into the Advanced Setting of Windows Firewall to allow ICMP Echo Requests to get ping to work with the Windows machines. There is an animation on slide 15 showing how to do this along with written directions on their handout (Question 11). Question 13 has them retrace the steps they would take to disable these firewall rules when finished pinging. Disabling the rules isn’t part of the lab, but it could be if you want them to do it.

Part 5: Tracing Tracks

- The tracert (Windows) and traceroute (Ubuntu) commands show the path that data takes through the network. Each “hop” represents a router or gateway the packet passes through, and the tool maps the route by sending packets using UDP.
- Installing traceroute on Ubuntu gives students hands-on experience with package installation and reinforces that not all tools are pre-installed on every system.
- In the cyber range, most traceroutes between machines will show just one hop since the machines are on the same local network segment.
- Traceroute from Ubuntu to Windows often fails unless the Windows firewall is completely disabled. This is because Linux traceroute uses UDP, and Windows Firewall blocks unknown inbound traffic by default. A fix isn’t as simple as enabling ICMP Echo Request (like with ping), because the traceroute packets target high, random UDP ports – and there’s no rule that can predict or allow all of them without opening up the firewall entirely.
- Turning off the firewall isn’t a great idea and isn’t necessary since students can still see the one hop from their Ubuntu machine to their partner’s and the one hop from their Windows machine to their partner’s.
- The **real-world traceroute to Google or 8.8.8.8** helps students compare simple internal routes to complex internet paths. They’ll see a mix of private **IPs (10.x.x.x, 192.168.x.x)** and **public IPs**, which shows the data moving through school or ISP networks before hitting the open internet.
- This is a good point to remind students that **longer paths don’t always mean worse connections** – but each hop adds delay, and failed hops can signal trouble spots.

Putting It All Together

- Why might a computer have an IP address but still be unreachable on a network?

A computer can have a valid IP address but still be unreachable if a firewall blocks incoming traffic (like ping) or if the service you're trying to access isn't running. In the lab, this happened with Windows 7 – pings failed until we enabled ICMP traffic in the firewall.

- What can open ports tell you about the services running on a machine? Why is this important for both troubleshooting and security?

Open ports show which services are listening for network traffic (e.g., port 80 for web traffic, port 22 for SSH). This is helpful for troubleshooting because it confirms which services are active. It's also a security concern – if unnecessary ports are open, they could be exploited by attackers.

- What did you learn from using both ping and traceroute? How do these tools help you understand what's happening between devices on a network?

Ping tests whether two devices can reach each other and shows packet loss. Traceroute maps the path data takes and shows how many hops are involved. Together, they help us understand both reachability and the route data travels. For example, the range machines usually have 1 hop, while internet traceroutes show multiple.

- How does DNS make using the internet easier for humans, and what did the nslookup results show you about how it actually works behind the scenes?

DNS translates domain names (like cyber.org) into IP addresses so humans don't have to remember long strings of characters. nslookup showed the actual IP addresses behind common websites and demonstrated how your machine asks a DNS server for help finding them.

Putting It All Together (10 minutes)

- Say to the students: “Today you learned how devices talk to each other across a network using protocols, IP addresses, MAC addresses, and ports. You explored how services like DNS and DHCP work behind the scenes to get data where it needs to go, and how routers, switches, and access points all play a role in making network communication possible. These basics are the building blocks of how we browse, email, stream, and connect every day.”
- Why is it important for devices on a network to follow the same protocols?

Because without shared protocols, devices wouldn’t understand each other. Protocols like TCP/IP and HTTP make sure data is sent, received, and processed correctly, no matter the device or operating system.
- How does knowing your IP and MAC address help with troubleshooting network problems?

If I know my IP and MAC address, I can check whether my device is getting the right network settings. It helps me figure out if the problem is with my device, the router, or the network setup.
- What’s one networking concept from today that you didn’t know before but now feel more confident about? Explain why it matters.

Answers will vary, but a sample answer could look like: “I didn’t know what NAT was before, but now I understand it’s how multiple devices can share one internet connection. It matters because it helps keep networks organized and saves IP addresses.”
- Say to students: “Every time you browse, stream, send, or click, your device follows a series of invisible steps to get data where it needs to go – fast, securely, and in the right format. Now that you understand those steps, you’re not just using the network... you’re starting to think like someone who could build or fix it.”